Swati Aggrawal

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PROFESSIONAL SUMMARY

MSc Computer Science graduate specializing in Artificial Intelligence and Machine Learning with hands-on experience building NLP applications, transformer-based models, AI experimentation, and model evaluation. Experienced in Python, SQL, PyTorch, Scikit-learn, FastAPI, Git, and generative AI workflows. Published IEEE researcher with practical experience in prompt engineering, semantic search, AI model benchmarking, and developing AI-powered applications. Passionate about applying AI to solve real-world business problems while continuously exploring emerging AI technologies. Enjoy experimenting with AI systems, evaluating model behavior, and understanding why models succeed or fail in practical applications.

EDUCATION

Sapienza Università di Roma **May 2026**

Master of Science in Computer Science

- **Relevant Coursework:** Deep Learning and Applied AI, NLP, Computer Vision, Distributed Systems.
- **Thesis:** Evaluating Pretrained Vision Transformers Under Continual Learning.
- Selected for a competitive international master's program.

Sharda University **July 2023**

Bachelor of Technology - CSE with AI and ML

- **Relevant Coursework:** Operating Systems, Data Structures, Machine Learning, Databases, Cloud Computing
- **Thesis:** Virtual Generative Model Inducing Human Identity to Computers in Metaverse
- Awarded **merit-based scholarship** (Top 5–10% of cohort).

TECHNICAL SKILLS

- **Programming Languages:** Python, SQL, C++, Bash
- **Artificial Intelligence & Machine Learning:** Machine Learning, Deep Learning, Natural Language Processing (NLP), Computer Vision, Transformer Models, Large Language Models (LLMs), Prompt Engineering, Model Evaluation, Cross Validation, Hyperparameter Tuning, Feature Engineering
- **Generative AI & NLP:** ChatGPT, Gemini, Claude, GitHub Copilot, Retrieval-Augmented Generation (RAG), Sentence Transformers, Embedding Models, Semantic Search, AI Workflow Design
- **Data Analysis & Visualization:** Pandas, NumPy, Statistical Analysis, Exploratory Data Analysis (EDA), Data Preprocessing, Data Visualization
- **Libraries & Frameworks:** Scikit-learn, PyTorch, TensorFlow, Keras, FastAPI
- **Deployment & Cloud:** REST APIs, Docker, AWS, API Integration, Workflow Automation
- **Version Control & Tools:** Git, GitHub, Linux

PROJECTS

AI Semantic Text Compression using Transformer Embeddings

- Developed a semantic text compression pipeline using GTR-T5 embeddings, vector quantization, and text reconstruction to preserve semantic information.
- Designed and conducted systematic experiments across multiple quantization levels to evaluate the trade-off between compression efficiency and reconstruction quality.
- Built reproducible experimentation pipelines in Python for data preprocessing, embedding generation, quantization, and performance evaluation.
- Performed data cleaning, preprocessing, and experimental analysis using **Python, Pandas, and NumPy**.
- Evaluated model performance using **Cosine Similarity** and **ROUGE-L** metrics to benchmark semantic reconstruction quality.
- Achieved optimal performance at **6-bit quantization**, obtaining a cosine similarity of **0.964** and a ROUGE-L score of **0.776**.
- Conducted quantitative analysis to identify non-linear relationships between compression levels and semantic reconstruction accuracy.

- Documented experimental findings, model outputs, and performance trade-offs to support reproducibility and future optimization.
- Strengthened understanding of transformer-based embeddings, NLP pipelines, AI experimentation, and model evaluation through practical implementation.
- **Technologies:** Python, GTR-T5, Sentence Transformers, Pandas, NumPy,Git

DocDelta [Document Comparison API]

- Developed an NLP-powered document intelligence system using transformer embeddings and semantic similarity techniques for automated comparison of legal and contract documents.
- Designed algorithms to identify added, removed, and modified content while distinguishing numerical changes from textual modifications.
- Built an AI-powered document comparison pipeline that automated semantic analysis of document revisions, significantly reducing manual review effort.
- Leveraged Sentence Transformers and embedding-based retrieval techniques to analyze semantic similarity between document versions.
- Deployed the application using FastAPI, Docker, AWS ECS Fargate, and CloudWatch for scalable inference and monitoring.
- Evaluated inference performance and optimized deployment architecture, achieving approximately 840ms warm inference latency.
- Designed prompt engineering workflows to improve semantic document comparison using transformer-based language models.
- Evaluated LLM outputs and refined prompts to improve consistency and semantic accuracy.

RESEARCH EXPERIENCE

Continual Learning for Computer Vision (Master's Thesis) — Nov 25 - May 26

- Designed and executed machine learning experiments to evaluate Vision Transformer performance across sequential learning tasks.
- Designed reproducible machine learning experimentation pipelines and comparative evaluation frameworks for transformer-based models.
- Conducted AI model benchmarking and comparative evaluation across multiple continual learning scenarios.
- Investigated catastrophic forgetting and knowledge retention in transformer-based visual learning systems.
- Automated experiment tracking and performance benchmarking across multiple continual learning scenarios.
- Conducted experiment tracking, performance analysis, and data processing using Pandas and NumPy.
- Documented experimental methodologies, findings, performance analyses, and future research opportunities to ensure reproducibility.
- **Technologies:** Python, PyTorch, Vision Transformers, Computer Vision

Machine Learning for Air Pollution [IEEE Publication] — May 2023

- Published an IEEE conference paper reviewing machine learning approaches for air quality prediction.
- Conducted comparative statistical analysis of six machine learning models using multiple performance metrics across environmental datasets.
- Evaluated prediction-based and knowledge-discovery methodologies across six NAAQS pollutants.
- Identified real-time monitoring and IoT integration as important future directions for environmental analytics.